BRAIN TUMOR CLASSIFICATION USING CNNS

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Abstract

Deep learning, particularly Convolutional Neural Networks (CNNs), has become a pivotal tool in medical image classification. Accurate and efficient brain tumor classification through MRI scans is critical for timely diagnosis and treatment planning. This study explores the application of two different CNN architectures: a standard CNN trained from scratch and a transfer learning model utilizing the EfficientNetB0 base. The primary objective is to classify brain tumors into four categories: glioma tumor, pituitary tumor, meningioma tumor, or no tumor. The CNN trained from scratch achieved an accuracy of 74%, whereas the EfficientNetB0 model, trained on the extensive database ImageNet, attained a higher accuracy of 96%. This comparison underscores the advantages of transfer learning in medical imaging, highlighting its potential for improving diagnostic outcomes.

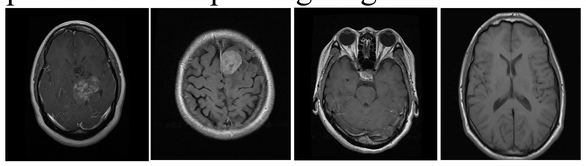


Figure 1. MRI scans of glioma, meningioma, pituitary, and no tumor (left to right).

Background

Traditional medical classification methods relied on manual interpretation, which was time-consuming and prone to human error. CNNs have shown significant promise in capturing patterns in medical images, facilitating early diagnosis, as demonstrated in several studies (Cai et al., 2020; Sarraf et al., 2017), where the LeNet-5 CNN framework to train Alzheimer's disease samples for sMRI and fMRI, achieving 98.84% and 96.85% accuracy. Furthermore, Kim et al. (2022) explored the efficacy of transfer learning, a method involving knowledge transfer across related tasks to improve performance on a new task. This method has overcame problems such as data, time, and hardware scarcity.

Method

We sourced a dataset from Kaggle, comprising 3264 MRI images divided into training and testing sets.
Preprocessing included image resizing, normalization, data augmentation, and categorical to improve model generalization.
Both models were evaluated using accuracy, precision, recall, and F1-score, with particular attention to overfitting and generalization capabilities.

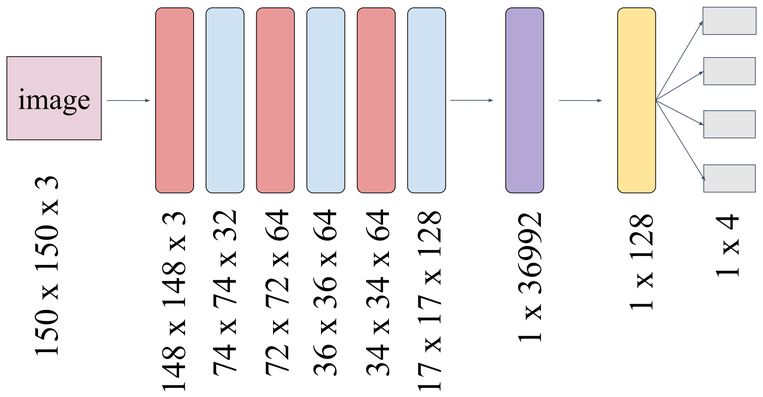


Figure 2. A visual representation of the architecture from the CNN from scratch

- Input layer: images' height and width are resized to 150x150 pixels, with 3 color channels (RGB)
- Conv2D layer: The convolution operation with a (3,3) filter reduces the spatial dimensions over time, applying an increasing number of filters (32, 64, 128). They use relu activation functions.
- Maxpooling2D layer: The layers reduce the spatial dimensions by half after each convolution operation while retaining the number of filters.
- Flatten layer: This converts the 3D tensor (17x17x128) into a 1D vector with 36,992 elements.
- Dense Layer: This outputs a vector with 128 elements.
- Output Dense Layer: A dropout of 0.5 is applied, and then final layer outputs a vector of 4 elements, corresponding to the probabilities of each tumor class

Results

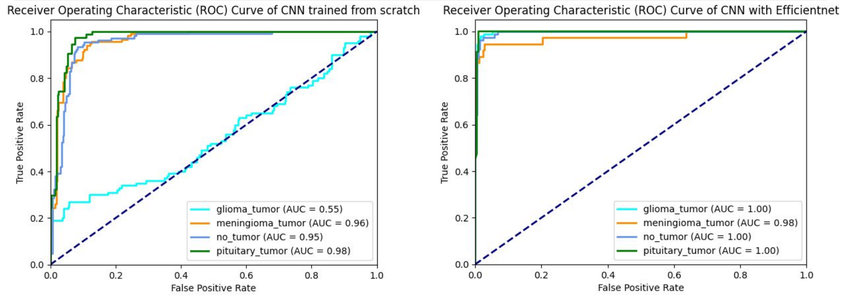


Figure 3, 4. ROC graph and AUC values of predictions made by both CNNs, from scratch and with EfficientB0 base

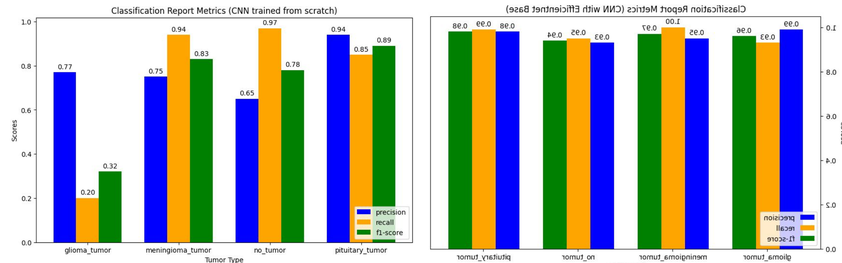


Figure 4, 5. Evaluation metrics values of both CNNs, from scratch and with EfficientB0 base

Discussion

- CNN from scratch: The Glioma class scored an AUC value of 0.55, compared 0.95 - 0.98 on other classes, barely better than random guessing.
- It's sensitivity/recall (0.17) was significantly lower. Several small factors could have contributed to the model's tendency to overfit on that particular class.
- The recall scores indicative of many false negative predictions due to the limitations of a simpler model, . The CNN with the EfficientNet base, on the other hand, demonstrated a superior performance achieving 98% classification accuracy in training and 96% accuracy on the test set.
- The high accuracy and reliability of the EfficientNet model highlight transfer learnings' potential utility in clinical settings for brain tumor diagnosis and medical image classification